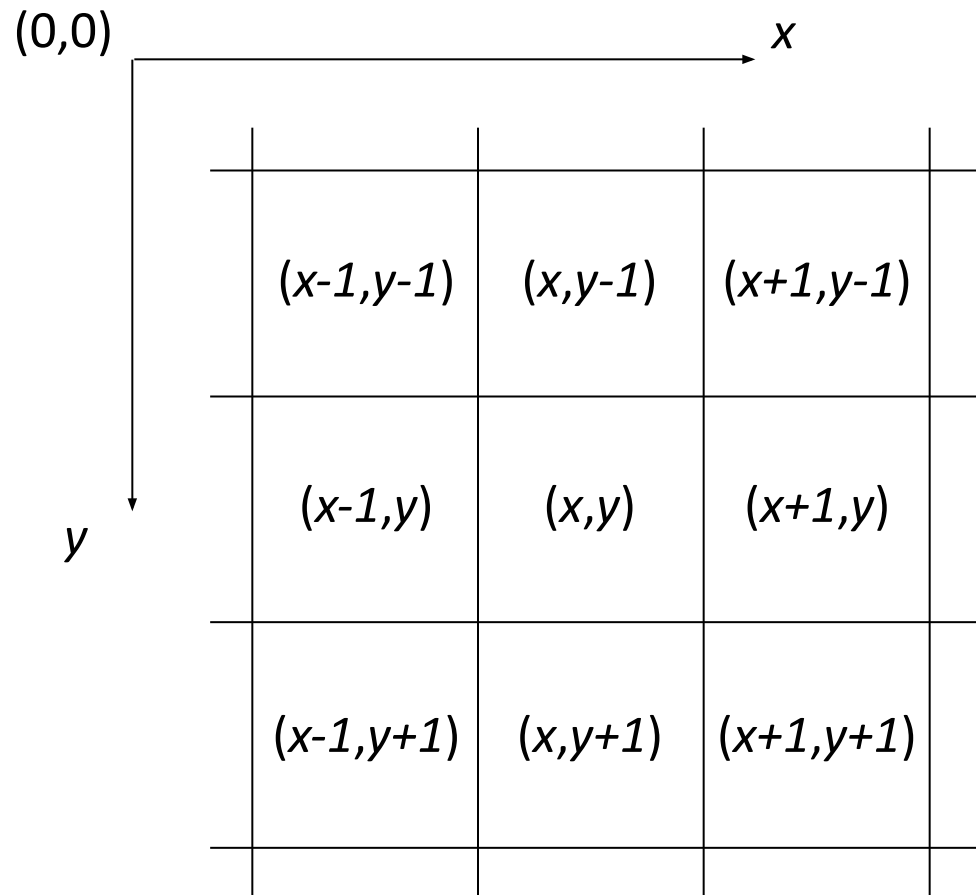


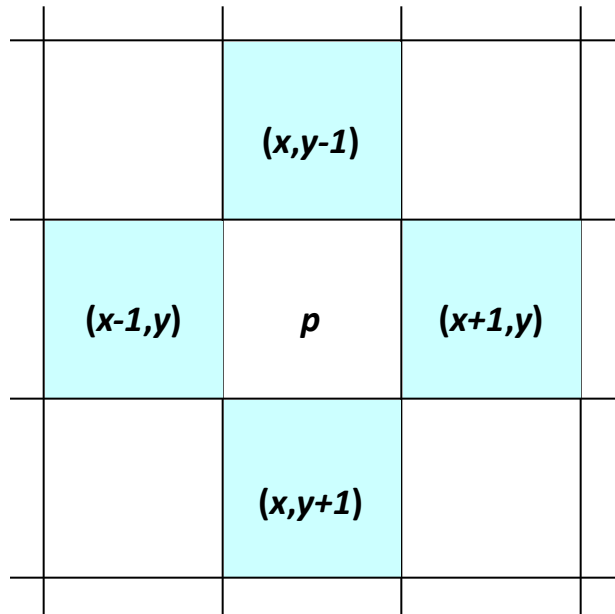
Basic Relationship of Pixels



Conventional indexing method

Neighbors of a Pixel

Neighborhood relation is used to tell adjacent pixels. It is useful for analyzing regions.



4-neighbors of p :

$$N_4(p) = \left\{ \begin{array}{l} (x-1,y) \\ (x+1,y) \\ (x,y-1) \\ (x,y+1) \end{array} \right\}$$

4-neighborhood relation considers only vertical and horizontal neighbors.

Note: $q \in N_4(p)$ implies $p \in N_4(q)$

Neighbors of a Pixel (cont.)

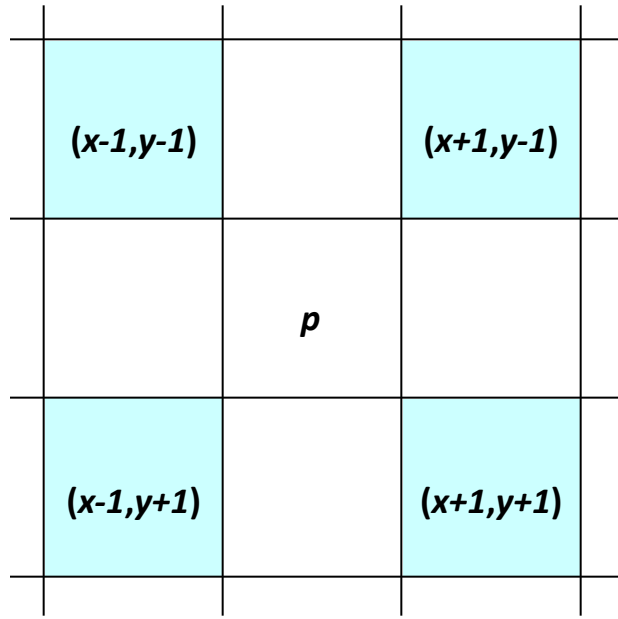
$(x-1, y-1)$	$(x, y-1)$	$(x+1, y-1)$
$(x-1, y)$	p	$(x+1, y)$
$(x-1, y+1)$	$(x, y+1)$	$(x+1, y+1)$

8-neighbors of p :

$$N_8(p) = \left\{ \begin{array}{l} (x-1, y-1) \\ (x, y-1) \\ (x+1, y-1) \\ (x-1, y) \\ (x+1, y) \\ (x-1, y+1) \\ (x, y+1) \\ (x+1, y+1) \end{array} \right\}$$

8-neighborhood relation considers all neighbor pixels.

Neighbors of a Pixel (cont.)



Diagonal neighbors of p :

$$N_D(p) = \left\{ \begin{array}{l} (x-1, y-1) \\ (x+1, y-1) \\ (x-1, y+1) \\ (x+1, y+1) \end{array} \right\}$$

Diagonal -neighborhood relation considers only diagonal neighbor pixels.

Connectivity

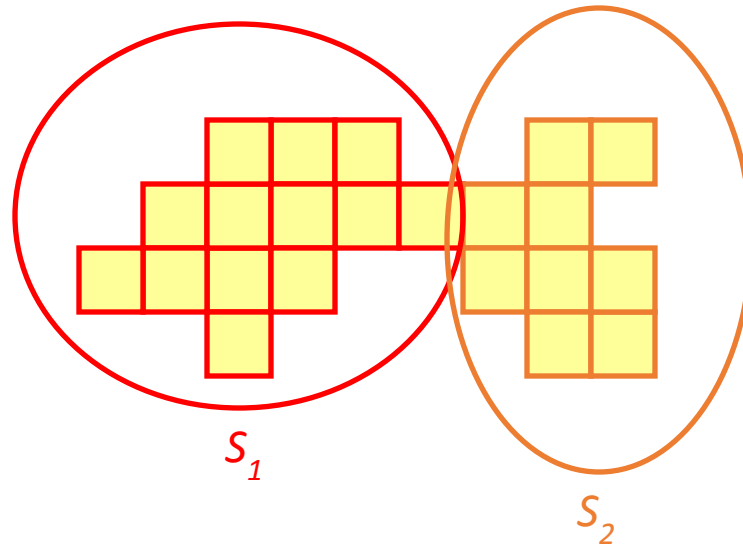
Connectivity is adapted from neighborhood relation. Two pixels are connected if they are in the same class (i.e. the same color or the same range of intensity) and they are neighbors of one another.

For p and q from the same class

- ◆ **4-connectivity**: p and q are 4-connected if $q \in N_4(p)$
- ◆ **8-connectivity**: p and q are 8-connected if $q \in N_8(p)$
- ◆ **mixed-connectivity (m-connectivity)**:
 p and q are m-connected if $q \in N_4(p)$ or
 $q \in N_D(p)$ and $N_4(p) \cap N_4(q) = \emptyset$

Adjacency

A pixel p is *adjacent* to pixel q if they are connected.
Two image subsets S_1 and S_2 are adjacent if some pixel in S_1 is adjacent to some pixel in S_2 .



We can define type of adjacency: 4-adjacency, 8-adjacency or m-adjacency depending on type of connectivity.

Path

A **path** from pixel p at (x,y) to pixel q at (s,t) is a sequence of distinct pixels:

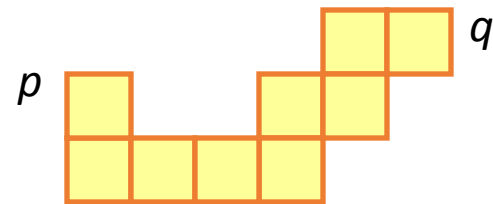
$$(x_0, y_0), (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

such that

$$(x_0, y_0) = (x, y) \text{ and } (x_n, y_n) = (s, t)$$

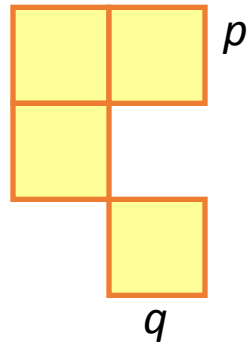
and

$$(x_i, y_i) \text{ is adjacent to } (x_{i-1}, y_{i-1}), \quad i = 1, \dots, n$$



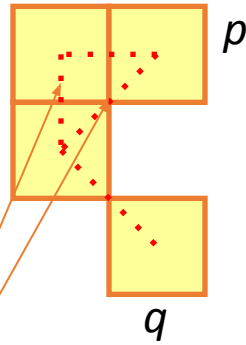
We can define type of path: 4-path, 8-path or m-path depending on type of adjacency.

Path (cont.)

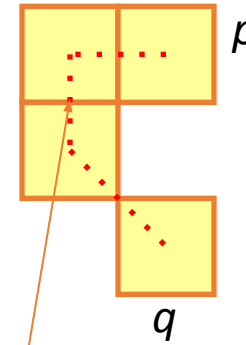


8-path from p to q
results in some ambiguity

8-path



m-path



m-path from p to q
solves this ambiguity

Distance

For pixel p , q , and z with coordinates (x,y) , (s,t) and (u,v) , D is a **distance function** or **metric** if

- ♦ $D(p,q) \geq 0$ ($D(p,q) = 0$ if and only if $p = q$)
- ♦ $D(p,q) = D(q,p)$
- ♦ $D(p,z) \leq D(p,q) + D(q,z)$

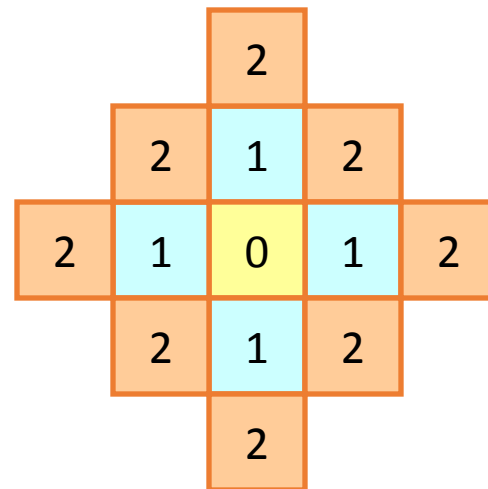
Example: Euclidean distance

$$D_e(p,q) = \sqrt{(x-s)^2 + (y-t)^2}$$

Distance (cont.)

D_4 -distance (city-block distance) is defined as

$$D_4(p, q) = |x - s| + |y - t|$$



Pixels with $D_4(p) = 1$ is 4-neighbors of p .

Distance (cont.)

D_8 -distance (chessboard distance) is defined as

$$D_8(p, q) = \max(|x - s|, |y - t|)$$

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Pixels with $D_8(p) = 1$ is 8-neighbors of p .